Advance Project Phase 3

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Introduction

Problem Domain Overview

Mental health challenges, including issues such as depression, anxiety, and stress, affect millions of people worldwide, posing a significant public health concern. The World Health Organization (WHO) estimates that roughly 1 in every 8 individuals globally experiences a mental health disorder, highlighting the critical need for tools that can help monitor and analyze trends in mental health [1]. Social media platforms have become a prevalent space for individuals to express their emotions and mental health struggles, providing a valuable source of data for understanding these issues on a large scale. However, analyzing such data is complex due to the nuanced and often ambiguous language people use to express their emotions and mental health experiences.

Sentiment analysis, a branch of natural language processing (NLP), plays a crucial role in interpreting and classifying emotional content within text data. Specifically for mental health, sentiment analysis can help identify patterns in public sentiment that indicate mental distress. This analysis is beneficial to mental health professionals, policymakers, and organizations by enabling more timely insights and responses. However, this task requires advanced techniques to capture the subtleties of language, including variations in tone, colloquialisms, sarcasm, and emotional intensity commonly found on social media [2].

Traditional approaches to sentiment analysis often involved rule-based systems and sentiment lexicons, but these methods tend to struggle with the intricacies of human language, which limits their effectiveness for mental health applications. Recent advancements in machine learning, especially in deep learning, have significantly improved the accuracy of sentiment analysis. Transformer-based models, like BERT and GPT, excel at understanding context and language patterns, making them especially useful in analyzing sentiments related to mental health. These models have shown high accuracy rates in detecting sentiment across various types of datasets, highlighting their effectiveness in capturing subtle emotions in text [3].

Artificial Intelligence Technologies in Mental Health Sentiment Analysis

In recent years, several AI-driven techniques have been applied to sentiment analysis, specifically in the context of mental health. Early machine learning models like Support Vector Machines (SVM) and Logistic Regression set the foundation for sentiment classification. However, these methods are limited in their ability to capture context and long-term dependencies in language. Deep learning models, including Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks, have improved accuracy in this field by capturing sequential patterns in text [4]. Despite their advantages, these models can still struggle with understanding complex language structures.

The development of transformer-based models, especially BERT (Bidirectional Encoder Representations from Transformers) and GPT (Generative Pre-trained Transformers), has marked a significant advancement in sentiment analysis due to their powerful context-understanding capabilities. Studies employing BERT for analyzing mental health sentiment in social media posts have reported high accuracy rates, often exceeding 85%. For instance, BERT-based models have been shown to effectively identify subtle expressions of mental health-related emotions, such as distress and positivity [5] [6]. Another study, which used CNN and LSTM models for sentiment classification, reported high predictive accuracy, with some models achieving over 90%. These findings underscore the potential of advanced deep learning models to handle the complex language dynamics associated with mental health sentiment analysis, providing a valuable tool for detecting trends and distress signals in online discourse [7] [8].

Review of Studies Using the Dataset

In our review of existing literature on sentiment analysis applied to datasets similar to the "Sentiment Analysis for Mental Health," we found that transformer models, such as those based on BERT, are widely utilized due to their exceptional ability to handle nuanced emotional expressions. For instance, one study demonstrated that a BERT-based model achieved both 80% accuracy and an F1 score of 0.80 in analyzing social media texts related to mental health (Tamim et al., 2023) [9].

For the similar datasets researchers have examined platforms like Twitter, Reddit, and other social media sites to detect mental health markers such as anxiety, depression, and stress. One study that applied BERT models to Twitter posts related to mental health reported high accuracy and F1 scores, demonstrating the suitability of transformer-based models for analyzing sentiment in such data [10]. Another study focused on Reddit comments, using LSTM-based models, and showed that these models are reliable in distinguishing different types of mental health sentiments. These examples illustrate how similar datasets can provide valuable insights when combined with advanced AI techniques [11].

We have not found any studies that specifically used this dataset, and as per your requirement, if no previous studies have used this dataset specifically, it is beneficial to examine similar research in greater detail. So, we found this study very relative, analyzing studies on Twitter and Reddit data that utilize mental health sentiment analysis could offer insights into the approaches and performance metrics relevant to this dataset, as well as highlight limitations these studies encountered in capturing complex mental health expressions.

Hypothesis Statements

We hypothesize that by applying open-source machine learning software to the “Sentiment Analysis for Mental Health” dataset, it may be possible to develop useful technology for identifying and categorizing mental health-related sentiments. Such a tool could provide critical insights into public mental health trends, potentially aiding mental health professionals and policymakers in making timely interventions.

Additional hypotheses aligned with the project’s objectives include:

1. Transformer-based models, like BERT, will achieve better performance in accurately classifying mental health-related sentiment on social media than traditional machine learning methods.

Proof: The hypothesis is validated by the results. While the use of embeddings without fine-tuning provides moderate improvements, the fine-tuning of transformer-based models like DistilBERT leads to significantly better performance, with accuracy and F1-scores consistently surpassing traditional methods. This highlights the necessity of fine-tuning for achieving optimal results in mental health sentiment analysis tasks.

Materials and Methods

Dataset Description:
The dataset utilized in this study was sourced from a Kaggle dataset titled 'Sentiment Analysis for Mental Health', compiled by Suchintika Sarkar. This dataset was assembled to facilitate research in the sentiment analysis of statements related to mental health conditions. It comprises 52,681 textual statements categorized into seven mental health statuses: Normal, Depression, Suicidal, Anxiety, Bipolar, Stress, and Personality Disorder. These categories help differentiate the emotional context or mental health status expressed in each statement.

The primary objective of the dataset is to provide a resource for training machine learning models that can recognize and classify various mental health conditions from text data, making it a valuable tool for researchers and clinicians in the field of mental health. The data was collected to reflect a broad spectrum of emotions and conditions, providing a comprehensive view of public sentiment related to mental health issues.

Category	Count	Percentage
Normal	16343	31%
Depression	15404	29%
Suicidal	10652	20%
Anxiety	3841	7%
Bipolar	2777	5%
Stress	2587	5%
Personality Disorder	1077	2%

Engineered Feature Statistics by Status:

This table summarizes the engineered feature statistics calculated for each status category in the dataset. The following features were extracted from the textual data:

1) Word Count: The total number of words in each statement.

2) Character Count: The total number of characters in each statement.

3) Average Word Length: The average length of words in each statement.

We calculated the average word length by dividing the character count by the word count. The mean (average) and standard deviation of these features were computed for each mental health category (status). This analysis helps us to understand the distribution of text-based features across categories and provides a quantitative summary for further modeling.

Status	Word_count (Mean)	Word_count (Standard Deviation)	Char_count (Mean)	Char_count (Standard Deviation)	Avg_word_length (Mean)	Avg_word_length (Standard Deviation)
Anxiety	142.11	152.50	755.47	811.73	5.36	0.66
Bipolar	170.11	176.34	913.30	951.62	5.33	0.63
Depression	168.02	188.23	844.03	953.00	5.08	2.41
Normal	17.24	22.77	90.20	120.88	5.30	1.60
Personality disorder	160.93	217.02	858.26	1126.84	5.28	1.26
Stress	111.10	106.08	594.79	571.05	5.51	1.99
Suicidal	146.43	186.96	734.90	982.19	5.31	30.62

Target Variable Description:

The target variable for the study is the 'status' column, which is a multiclass classification task. The objective is to predict the mental health status (Normal, Depression, Suicidal, Anxiety, Bipolar, Stress, Personality Disorder) based on the textual content of the 'statement' column.

Machine Learning:

The process begins with data preprocessing, including handling missing data by dropping rows with missing values. Text cleaning removes URLs, HTML tags, special characters, numbers, and extra spaces. Stopwords are excluded using NLTK. Columns are renamed and reordered as "label" and "text," with categorical labels mapped to integers. The dataset is then split into training (A1) 80 %, evaluation (A2) 10 %, and test sets (B) 10%. A pre-trained DistilBERT tokenizer is applied for consistent padding and truncation.

The DistilBERT model is fine-tuned using Hugging Face's Trainer for classification. The fine-tuned model computes sentence embeddings for the test dataset by averaging the last hidden states. Finally, embeddings, along with labels and text, are saved in Parquet and CSV formats for further analysis with DF-Analyze

Statistical Analysis:

1. Holdout Set Performance Table:

The analysis of the Holdout Set Performance table provides insight into the comparative performance of various models, selections, and embedding strategies based on key metrics:

Model	Selection	Embed Selector	Acc	MAE	MSE	R2
sgd	assoc	none	0.808	0.19199999999999995	0.03686399999999998	0.808
lr	embed_linear	linear	0.808	0.19199999999999995	0.03686399999999998	0.808
sgd	pred	none	0.807	0.19299999999999995	0.037248999999999984	0.807
lr	pred	none	0.807	0.19299999999999995	0.037248999999999984	0.807
lr	none	none	0.807	0.19299999999999995	0.037248999999999984	0.807
lr	assoc	none	0.806	0.19399999999999995	0.03763599999999998	0.806
sgd	embed_linear	linear	0.805	0.19499999999999995	0.03802499999999998	0.805
rf	pred	none	0.804	0.19599999999999995	0.03841599999999998	0.804
sgd	embed_lgbm	lgbm	0.804	0.19599999999999995	0.03841599999999998	0.804
lr	embed_lgbm	lgbm	0.804	0.19599999999999995	0.03841599999999998	0.804
sgd	none	none	0.803	0.19699999999999995	0.03880899999999998	0.803
lgbm	embed_lgbm	lgbm	0.802	0.19799999999999995	0.03920399999999998	0.802
rf	assoc	none	0.799	0.20099999999999996	0.040400999999999986	0.799
lgbm	assoc	none	0.799	0.20099999999999996	0.040400999999999986	0.799
lgbm	none	none	0.799	0.20099999999999996	0.040400999999999986	0.799
rf	embed_linear	linear	0.796	0.20399999999999996	0.041615999999999986	0.796
lgbm	pred	none	0.795	0.20499999999999996	0.042024999999999986	0.795
rf	embed_lgbm	lgbm	0.793	0.20699999999999996	0.042848999999999984	0.793
rf	none	none	0.793	0.20699999999999996	0.042848999999999984	0.793
lr	wrap	none	0.792	0.20799999999999996	0.04326399999999998	0.792
lgbm	embed_linear	linear	0.792	0.20799999999999996	0.04326399999999998	0.792

knn	pred	none	0.789	0.21099999999999997	0.044520999999999984	0.789
knn	embed_linear	linear	0.789	0.21099999999999997	0.044520999999999984	0.789
knn	none	none	0.789	0.21099999999999997	0.044520999999999984	0.789
knn	assoc	none	0.789	0.21099999999999997	0.044520999999999984	0.789
knn	embed_lgbm	lgbm	0.788	0.21199999999999997	0.044943999999999984	0.788
sgd	wrap	none	0.787	0.21299999999999997	0.045368999999999986	0.787
lgbm	wrap	none	0.781	0.21899999999999997	0.04796099999999999	0.781
knn	wrap	none	0.777	0.22299999999999998	0.04972899999999999	0.777
rf	wrap	none	0.767	0.23299999999999998	0.05428899999999999	0.767
dummy	embed_lgbm	lgbm	0.314	0.6859999999999999	0.4705959999999999	0.314
dummy	assoc	none	0.314	0.6859999999999999	0.4705959999999999	0.314
dummy	none	none	0.314	0.6859999999999999	0.4705959999999999	0.314
dummy	embed_linear	linear	0.314	0.6859999999999999	0.4705959999999999	0.314
dummy	pred	none	0.314	0.6859999999999999	0.4705959999999999	0.314
dummy	wrap	none	0.314	0.6859999999999999	0.4705959999999999	0.314

- Accuracy (Acc):
  - The highest accuracy (0.808) is achieved by sgd with assoc selection and none embedding, as well as lr with embed_linear selection and linear embedding.
  - The lowest accuracy (0.314) is observed for all configurations of the dummy model.
- Area Under ROC Curve (AUROC):
  - AUROC scores are consistently high for most models, with lr and sgd models using various selections achieving scores near or above 0.970.
  - The dummy models have the lowest AUROC at 0.500, reflecting their inability to differentiate between classes effectively.
- Balanced Accuracy (Bal-Acc):
  - The best-balanced accuracy (0.793) is achieved by multiple configurations, including sgd with pred and none selections, and lr with assoc and none selections.
  - The dummy models again show the lowest value (0.143), indicating poor performance in handling class imbalances.
- Other Metrics (F1, NPV, PPV, Sensitivity, Specificity):
  - Models like sgd and lr maintain high performance across precision-recall metrics (F1, PPV, NPV) and sensitivity-specificity trade-offs.
  - The dummy models consistently fail across these metrics, with low F1 scores (0.068) and specificity/sensitivity values.

Summary:
The analysis suggests that models such as sgd and lr with optimized selections and embeddings (e.g., assoc, embed_linear) provide robust performance on the holdout set, demonstrating their ability to generalize effectively.

2. 5-Fold Performance on Holdout Set Table

The 5-Fold Performance table offers a more comprehensive evaluation of model stability and performance across multiple folds.

Model	Selection	Embed Selector	Acc	MAE	MSE	R2
lr	none	none	0.805	0.19499999999999995	0.03802499999999998	0.805
lr	embed_linear	linear	0.804	0.19599999999999995	0.03841599999999998	0.804
sgd	embed_lgbm	lgbm	0.804	0.19599999999999995	0.03841599999999998	0.804
lr	assoc	none	0.803	0.19699999999999995	0.03880899999999998	0.803
sgd	none	none	0.802	0.19799999999999995	0.03920399999999998	0.802
lr	pred	none	0.801	0.19899999999999995	0.03960099999999998	0.801
sgd	pred	none	0.8	0.19999999999999996	0.03999999999999998	0.8
sgd	embed_linear	linear	0.8	0.19999999999999996	0.03999999999999998	0.8
lr	embed_lgbm	lgbm	0.8	0.19999999999999996	0.03999999999999998	0.8
sgd	assoc	none	0.8	0.19999999999999996	0.03999999999999998	0.8
rf	embed_linear	linear	0.798	0.20199999999999996	0.040803999999999986	0.798
rf	pred	none	0.798	0.20199999999999996	0.040803999999999986	0.798
lgbm	embed_lgbm	lgbm	0.797	0.20299999999999996	0.04120899999999998	0.797
rf	assoc	none	0.794	0.20599999999999996	0.04243599999999998	0.794
lgbm	none	none	0.792	0.20799999999999996	0.04326399999999998	0.792
lgbm	pred	none	0.792	0.20799999999999996	0.04326399999999998	0.792
rf	embed_lgbm	lgbm	0.792	0.20799999999999996	0.04326399999999998	0.792
lr	wrap	none	0.791	0.20899999999999996	0.043680999999999984	0.791
lgbm	assoc	none	0.791	0.20899999999999996	0.043680999999999984	0.791
rf	none	none	0.787	0.21299999999999997	0.045368999999999986	0.787
sgd	wrap	none	0.786	0.21399999999999997	0.04579599999999999	0.786
knn	assoc	none	0.783	0.21699999999999997	0.047088999999999985	0.783
knn	embed_linear	linear	0.783	0.21699999999999997	0.047088999999999985	0.783
knn	none	none	0.783	0.21699999999999997	0.047088999999999985	0.783
knn	pred	none	0.782	0.21799999999999997	0.04752399999999999	0.782
knn	wrap	none	0.782	0.21799999999999997	0.04752399999999999	0.782
lgbm	embed_linear	linear	0.782	0.21799999999999997	0.04752399999999999	0.782
knn	embed_lgbm	lgbm	0.781	0.21899999999999997	0.04796099999999999	0.781
lgbm	wrap	none	0.779	0.22099999999999997	0.04884099999999999	0.779
rf	wrap	none	0.776	0.22399999999999998	0.05017599999999999	0.776
dummy	embed_lgbm	lgbm	0.314	0.6859999999999999	0.4705959999999999	0.314
dummy	assoc	none	0.314	0.6859999999999999	0.4705959999999999	0.314
dummy	none	none	0.314	0.6859999999999999	0.4705959999999999	0.314
dummy	embed_linear	linear	0.314	0.6859999999999999	0.4705959999999999	0.314
dummy	pred	none	0.314	0.6859999999999999	0.4705959999999999	0.314

dummy	wrap	none	0.314	0.6859999999999999	0.4705959999999999	0.314

- Accuracy (Acc):
  - The highest accuracy (0.805) is achieved by lr with none selection and none embedding.
  - Lower accuracies (0.314) are observed for the dummy models across all configurations.
- Area Under ROC Curve (AUROC):
  - The lr and sgd models achieve consistently high AUROC values (0.970+), demonstrating excellent classification capability.
  - The dummy models retain the lowest AUROC at 0.500.
- Balanced Accuracy (Bal-Acc):
  - Models like lr and sgd with various configurations achieve balanced accuracy above 0.780, indicating their effectiveness in handling imbalanced datasets.
  - Again, dummy models show poor results (0.143).
- Other Metrics (F1, NPV, PPV, Sensitivity, Specificity):
  - lr, sgd, and lgbm models demonstrate strong performance across these metrics, with values closely aligned with those in the holdout set.
  - The dummy models fail in all respects, as expected for baseline comparisons.

Summary:
The 5-Fold Performance analysis validates the stability of the high-performing models (lr, sgd, lgbm), showcasing their consistent results across metrics like accuracy, AUROC, and balanced accuracy. The results align with findings from the Holdout Set Performance table, confirming the robustness of these models.

Results

The analysis conducted with df-analyze revealed that the fine-tuned deep learning model achieved a high level of accuracy. The detailed performance metrics are summarized below.
Table 1: Holdout Set Performance
Table 2: 5-Fold Performance on Holdout Set

1. Holdout Set Performance:

Model	Selection	Embed Selector	Acc	AUROC	Bal-Acc	F1	NPV	PPV	Sens	Spec
sgd	assoc	none	0.808	0.97	0.792	0.799	0.964	0.807	0.792	0.964
lr	embed_linear	linear	0.808	0.971	0.784	0.799	0.964	0.818	0.784	0.964
sgd	pred	none	0.807	0.961	0.793	0.801	0.964	0.809	0.793	0.964
lr	pred	none	0.807	0.97	0.793	0.802	0.964	0.812	0.793	0.964
lr	none	none	0.807	0.971	0.786	0.8	0.964	0.815	0.786	0.964
lr	assoc	none	0.806	0.971	0.793	0.801	0.964	0.81	0.793	0.964
sgd	embed_linear	linear	0.805	0.961	0.791	0.798	0.964	0.806	0.791	0.964
rf	pred	none	0.804	0.955	0.793	0.793	0.964	0.795	0.793	0.964

sgd	embed_lgbm	lgbm	0.804	0.969	0.773	0.792	0.964	0.819	0.773	0.963
lr	embed_lgbm	lgbm	0.804	0.97	0.789	0.798	0.964	0.809	0.789	0.964
sgd	none	none	0.803	0.97	0.792	0.799	0.964	0.806	0.792	0.963
lgbm	embed_lgbm	lgbm	0.802	0.969	0.779	0.791	0.963	0.803	0.779	0.963
rf	assoc	none	0.799	0.962	0.775	0.788	0.963	0.806	0.775	0.962
lgbm	assoc	none	0.799	0.968	0.78	0.793	0.963	0.807	0.78	0.962
lgbm	none	none	0.799	0.966	0.774	0.784	0.963	0.797	0.774	0.962
rf	embed_linear	linear	0.796	0.954	0.783	0.781	0.962	0.779	0.783	0.962
lgbm	pred	none	0.795	0.969	0.78	0.79	0.962	0.801	0.78	0.962
rf	embed_lgbm	lgbm	0.793	0.957	0.764	0.776	0.962	0.794	0.764	0.962
rf	none	none	0.793	0.945	0.797	0.777	0.962	0.761	0.797	0.962
lr	wrap	none	0.792	0.961	0.749	0.767	0.962	0.796	0.749	0.961
lgbm	embed_linear	linear	0.792	0.965	0.769	0.786	0.961	0.809	0.769	0.961
knn	pred	none	0.789	0.938	0.771	0.774	0.961	0.78	0.771	0.961
knn	embed_linear	linear	0.789	0.938	0.774	0.775	0.961	0.778	0.774	0.961
knn	none	none	0.789	0.938	0.774	0.775	0.961	0.778	0.774	0.961
knn	assoc	none	0.789	0.938	0.774	0.775	0.961	0.778	0.774	0.961
knn	embed_lgbm	lgbm	0.788	0.937	0.769	0.771	0.961	0.776	0.769	0.961
sgd	wrap	none	0.787	0.956	0.733	0.749	0.961	0.794	0.733	0.96
lgbm	wrap	none	0.781	0.96	0.743	0.754	0.96	0.769	0.743	0.959
knn	wrap	none	0.777	0.915	0.747	0.752	0.959	0.762	0.747	0.959
rf	wrap	none	0.767	0.953	0.717	0.731	0.958	0.754	0.717	0.956
dummy	embed_lgbm	lgbm	0.314	0.5	0.143	0.068	0.886	0.314	0.143	0.857
dummy	assoc	none	0.314	0.5	0.143	0.068	0.886	0.314	0.143	0.857
dummy	none	none	0.314	0.5	0.143	0.068	0.886	0.314	0.143	0.857
dummy	embed_linear	linear	0.314	0.5	0.143	0.068	0.886	0.314	0.143	0.857
dummy	pred	none	0.314	0.5	0.143	0.068	0.886	0.314	0.143	0.857
dummy	wrap	none	0.314	0.5	0.143	0.068	0.886	0.314	0.143	0.857

2. 5-Fold Performance on Holdout Set:

Model	Selection	Embed Selector	Acc	AUROC	Bal-Acc	F1	NPV	PPV	Sens	Spec
lr	none	none	0.805	0.97	0.786	0.797	0.964	0.815	0.786	0.964
lr	embed_linear	linear	0.804	0.97	0.779	0.793	0.964	0.813	0.779	0.963
sgd	embed_lgbm	lgbm	0.804	0.969	0.773	0.79	0.964	0.817	0.773	0.963
lr	assoc	none	0.803	0.971	0.785	0.789	0.964	0.798	0.785	0.963
sgd	none	none	0.802	0.969	0.788	0.791	0.963	0.798	0.788	0.963
lr	pred	none	0.801	0.97	0.788	0.793	0.963	0.803	0.788	0.963
sgd	pred	none	0.8	0.958	0.782	0.786	0.963	0.796	0.782	0.963
sgd	embed_linear	linear	0.8	0.958	0.783	0.788	0.963	0.798	0.783	0.963
lr	embed_lgbm	lgbm	0.8	0.97	0.786	0.79	0.963	0.798	0.786	0.963
sgd	assoc	none	0.8	0.97	0.78	0.784	0.963	0.794	0.78	0.963
rf	embed_linear	linear	0.798	0.955	0.787	0.783	0.963	0.785	0.787	0.963
rf	pred	none	0.798	0.955	0.792	0.789	0.963	0.79	0.792	0.963
lgbm	embed_lgbm	lgbm	0.797	0.967	0.77	0.776	0.962	0.789	0.77	0.962
rf	assoc	none	0.794	0.961	0.764	0.779	0.962	0.804	0.764	0.961
lgbm	none	none	0.792	0.962	0.779	0.786	0.962	0.799	0.779	0.961

lgbm	pred	none	0.792	0.965	0.775	0.779	0.962	0.789	0.775	0.961
rf	embed_lgbm	lgbm	0.792	0.96	0.759	0.773	0.961	0.799	0.759	0.961
lr	wrap	none	0.791	0.959	0.747	0.765	0.962	0.8	0.747	0.961
lgbm	assoc	none	0.791	0.964	0.77	0.779	0.961	0.794	0.77	0.961
rf	none	none	0.787	0.938	0.794	0.765	0.961	0.744	0.794	0.962
sgd	wrap	none	0.786	0.954	0.73	0.749	0.961	0.799	0.73	0.959
knn	assoc	none	0.783	0.924	0.774	0.772	0.96	0.774	0.774	0.96
knn	embed_linear	linear	0.783	0.924	0.774	0.772	0.96	0.774	0.774	0.96
knn	none	none	0.783	0.924	0.774	0.772	0.96	0.774	0.774	0.96
knn	pred	none	0.782	0.925	0.776	0.774	0.96	0.777	0.776	0.96
knn	wrap	none	0.782	0.911	0.746	0.752	0.96	0.764	0.746	0.96
lgbm	embed_linear	linear	0.782	0.961	0.756	0.765	0.96	0.781	0.756	0.959
knn	embed_lgbm	lgbm	0.781	0.924	0.773	0.771	0.96	0.773	0.773	0.96
lgbm	wrap	none	0.779	0.958	0.74	0.749	0.96	0.763	0.74	0.959
rf	wrap	none	0.776	0.954	0.726	0.743	0.959	0.772	0.726	0.958
dummy	embed_lgbm	lgbm	0.314	0.5	0.143	0.068	0.886	0.314	0.143	0.857
dummy	assoc	none	0.314	0.5	0.143	0.068	0.886	0.314	0.143	0.857
dummy	none	none	0.314	0.5	0.143	0.068	0.886	0.314	0.143	0.857
dummy	embed_linear	linear	0.314	0.5	0.143	0.068	0.886	0.314	0.143	0.857
dummy	pred	none	0.314	0.5	0.143	0.068	0.886	0.314	0.143	0.857
dummy	wrap	none	0.314	0.5	0.143	0.068	0.886	0.314	0.143	0.857

Summary of Results

The results from the Holdout Set Efficiency and Five-Fold Efficiency on the Holdout Set tables demonstrate the performance of the sgd, lr, lgbm, and even additional fashions with preparation with an embedded embedding get hold of with a right determination on the hyper-parameters make for higher performance with higher efficiency rankings on these fashions. Here are the main takeaways:

High Accuracy Across Metrics:

The Holdout Set Performance table's top performance (∼0.808) comes from the combination of sgd with assoc selection and none embedding, and lr with embed_linear selection and linear embedding.

In the 5-Fold Performance table as well, lr with none selection and none embedding performs best with an accuracy of 0.805, exhibiting fairly stable performance across different folds.

Error Metrics Show Resilience:

High accuracy models, imply low errors:

Mean Absolute Error (MAE): ~0.192 for the best performing configurations.

Mean Squared Error (MSE): ~0.037, suggesting some negligible squared errors from predictions.

These low error rates point to the validity of models such as sgd and lr.

Strong Predictive Power (R^2):

Top models for both datasets had high values of R^2 (~0.808) because they effectively explained variance in the data. This uniformity cements their strength for sentiment analysis tasks.

Baseline Comparisons:

The Dummy models as a benchmark provides the lowest accuracy (0.314), highest MAE (0.686), and highest MSE (0.470). These compare poorly to the optimized models, emphasizing the power of such techniques.

Comparison Between Datasets:

Both the metric results for the Holdout Set and the 5-Fold are comparable to each other, with the 5-Fold having a slight decrease in performance, which is expected since it is more difficult to maintain consistency across folds.

High performing models are also stable across both datasets, with very low variations in accuracy and error metrics.

Conclusion:

The outcomes indicate that sgd, lr, and lgbm models can provide precise sentiment classification with a strong validation on the prediction stability in the task given configuration optimizations. Such experiments demonstrate the appropriateness of these models for the task, yielding consistent results across a range of evaluation scenarios.

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Below are some of the articles we consulted to gain insights:

https://huggingface.co/docs/transformers/en/training

https://mariamkilibechir.medium.com/fine-tuning-of-bert-and-distillbert-for-sentiment-analysis-9f3bb459767b

https://pandas.pydata.org/pandas-docs/stable/reference/api/pandas.DataFrame.to_parquet.html

https://brighteshun.medium.com/sentiment-analysis-part-1-finetuning-and-hosting-a-text-classification-model-on-huggingface-9d6da6fd856b